

# TX4 Exact P4 / GFL11 Julia Reproduction

IAS 2026 independent validation bundle

2026-09-20

## Verdict

The exact frozen TX4 P4 flagship closes as Case A for the requested custom same-model Python-vs-Julia reproduction. The independent Julia implementation reproduces the frozen Python V4 census with the exact 11-state GFL, exact P4 SG policy, matched AC equilibrium, 16-case verdict agreement, and critical voltage-mode MAC of 0.9999999999999988 for H4.

This is an IAS poster-credibility upgrade for the stated same-model spectral claim. It is not an EMT, PowerDynamics SimpleGFLDC, or TPWRS-equivalence result.

## Frozen setup

- H4:  $\{30, 33, 35, 37\}$ .
- P4:  $(g, k, t, h) = (0.03625, 1.425, 1.5, 1.0)$ .
- GFL11 order:  $[\theta_{pll}, x_{pll}, p_f, q_f, x_p, x_q, i_d, i_q, x_{id}, x_{iq}, x_v]$ .
- Matched P/Q initialization; all 16 cases share the frozen AC point.
- The prior GFL10 harness remains separate and is not relabeled as exact.

## Primary gates

| Gate | Result                                                                                                                                 |
|------|----------------------------------------------------------------------------------------------------------------------------------------|
| G1   | <b>PASS</b> Device-level derivatives and injections match exactly at the fixed parity point; maximum difference is 0.0.                |
| G2   | <b>PASS</b> H4 has $86 = 4 \times 11 + 6 \times 7$ dynamic states.                                                                     |
| G3   | <b>PASS</b> All 16 Python/Julia stable-unstable verdicts are identical.                                                                |
| G4   | <b>PASS</b> H4 alpha: Python 0.1270064680506376, Julia 0.12700646832229723; difference $2.72 \times 10^{-10} \text{ s}^{-1}$ .         |
| G5   | <b>PASS</b> H4 frequency: Python 0.6222796695029233 Hz, Julia 0.6222796695187043 Hz; difference $1.58 \times 10^{-11} \text{ Hz}$ .    |
| G6   | <b>PASS</b> H4 voltage-mode MAC is 0.9999999999999988.                                                                                 |
| G7   | <b>PASS</b> Maximum voltage-magnitude difference $2.29 \times 10^{-14}$ ; maximum angle difference $2.30 \times 10^{-8} \text{ rad}$ . |
| G8   | <b>PASS</b> Equation, state-order, P4-policy, device-parity, and all-SG parity audits are present.                                     |

All 15 proper subsets are stable in both codes. H4 is unstable in both codes with the same positive transverse critical mode.

## State-count correction

The prior independent Julia H4 result had 82 states because it used GFL10. The exact reproduction has 86 states because four replaced converters each add  $x_v$ .

| Case           | States | Interpretation                             |
|----------------|--------|--------------------------------------------|
| Prior GFL10 H4 | 82     | retained as historical, non-exact evidence |
| Exact GFL11 H4 | 86     | this branch, exact P4/GFL11                |

## Validation figures

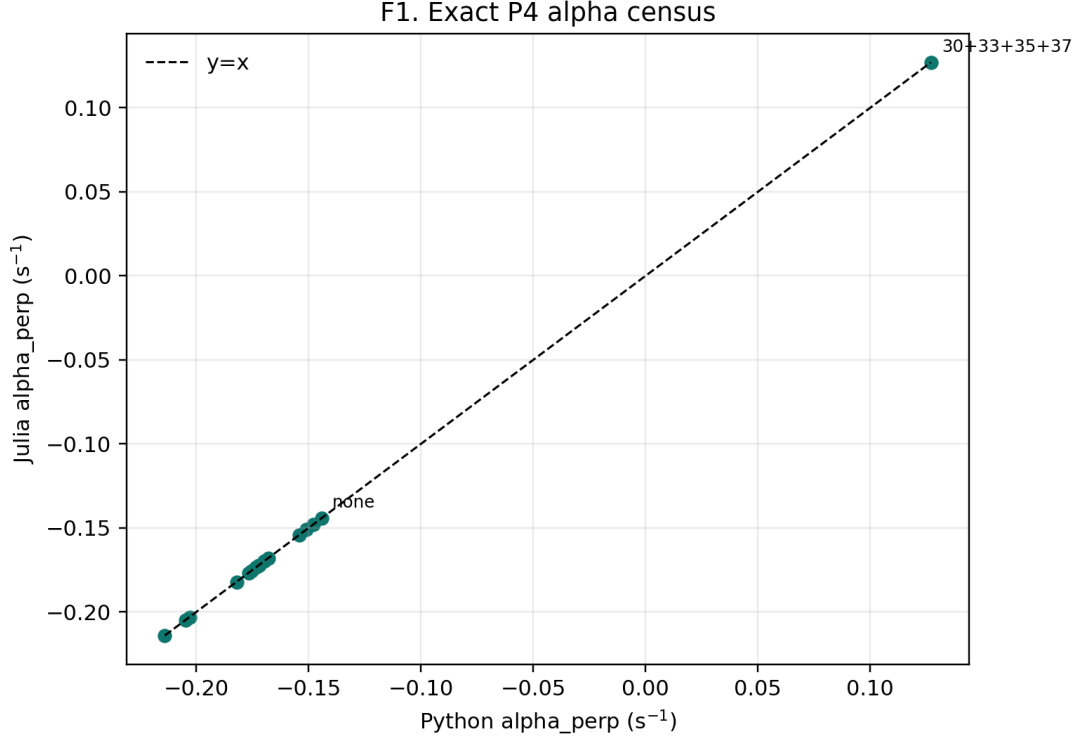


Figure 1: All 16 alpha values: Python versus Julia.

## Evidence and limitations

The complete numerical evidence is in the cross-code census, all-SG parity, device parity, mode-match, and state-count CSV files. The exact equations and policy are audited in the two accompanying Markdown documents.

The allowed claim is limited to cross-language reproduction of the frozen P4/GFL11 same-model spectral result for the declared reduced custom IEEE-39 DAE. It does not validate EMT behavior, a second independent dynamic model, controller saturation, protection, faults, or TPWRS submission readiness.

## Reproducibility

Branch: `research/tx4-exact-p4-julia-reproduction`. Parent: `069fa21204aa7c6f45fb44eb17e304ab99685a2b`. No push was performed.

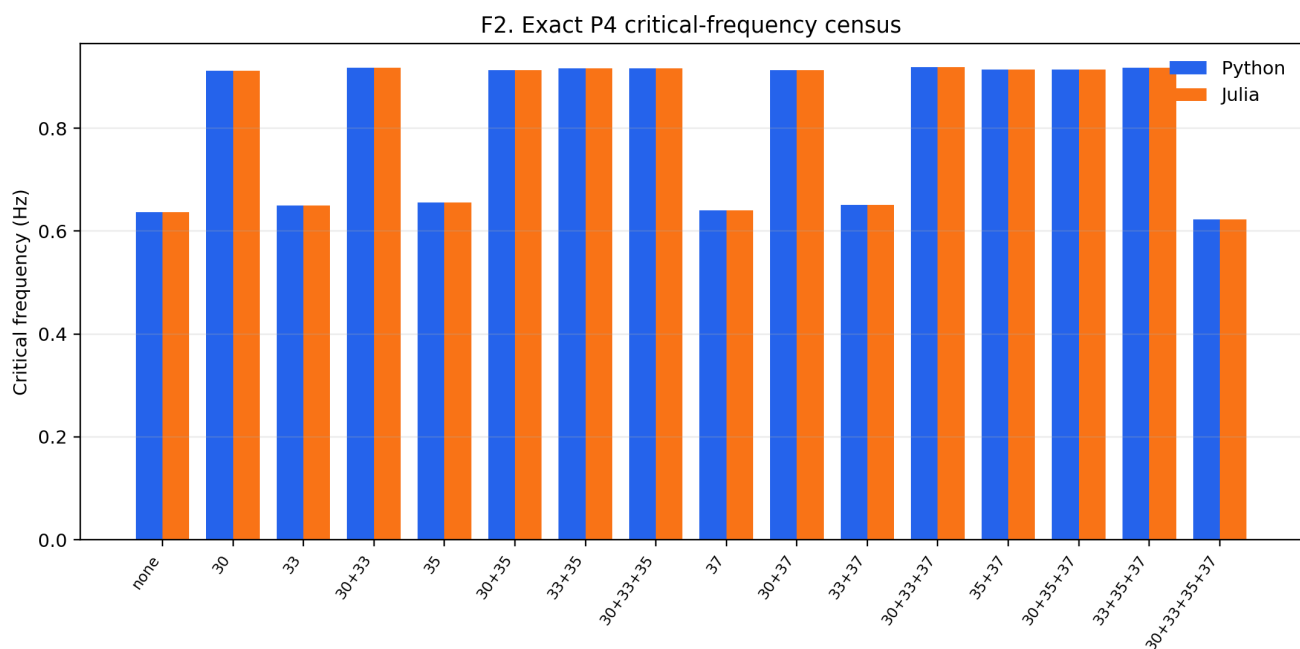


Figure 2: All 16 critical frequencies.

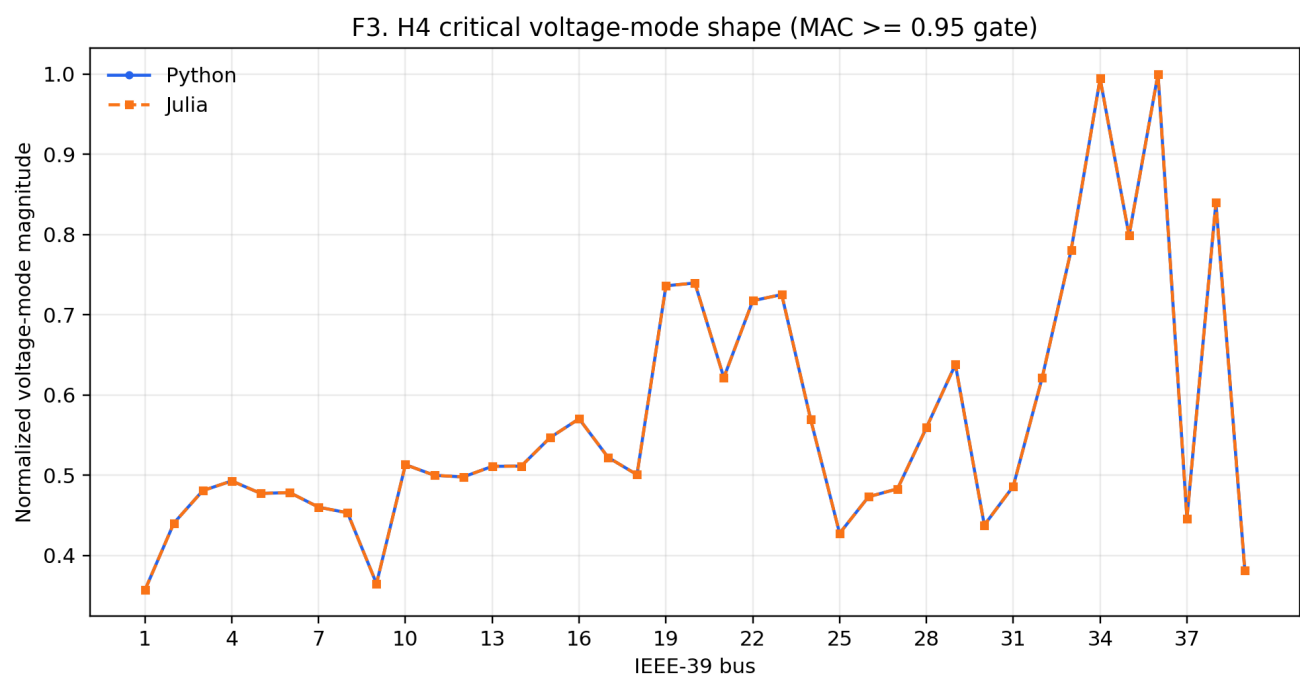


Figure 3: H4 critical voltage-mode magnitude by IEEE-39 bus.

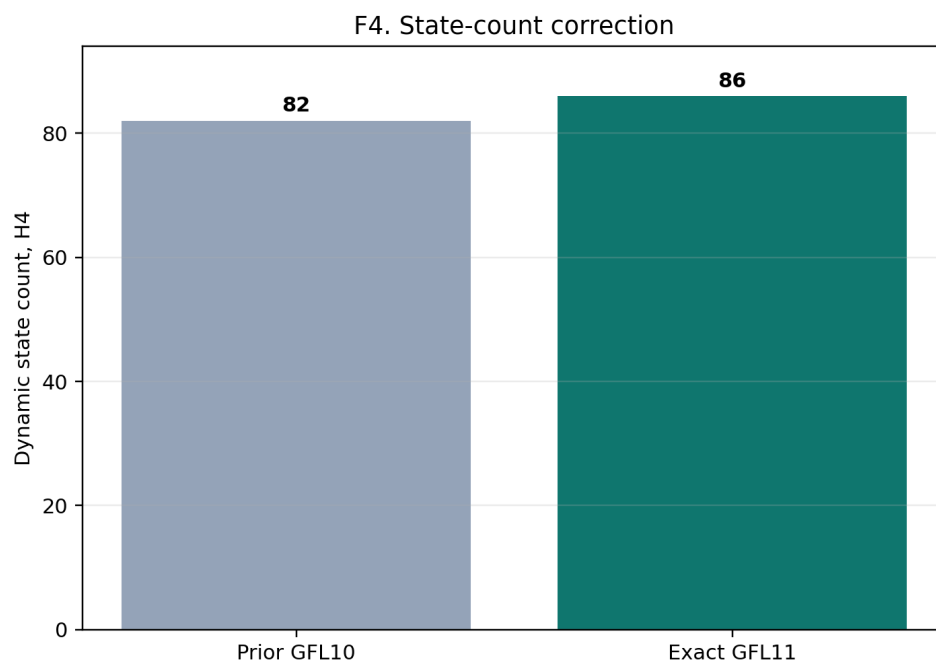


Figure 4: Correction from prior GFL10 to exact GFL11 state count.